■# PAPER<i>349 — SPT-CL J2215: Highest F</i>U<i>Bi</i>i in UQFF Dataset — Cool Core Starburst at z=1.16

Author: Daniel T. Murphy

SPT-CL J2215-3537 (z = 1.16, M = 7.32×10^{1}■ M■, SFR ≈ 700 M■/yr) is the most extreme cool-core cluster in the South Pole Telescope sample and yields the highest UQFF buoyancy-unified force in the entire PAPER346–352 dataset: FUBii ≈ -1.40×10^{21}■ N. The extreme starburst provides an independently measured SFR confirming the UQFF SFR = ρgas·vwind·fres formula. The x2 = 8.4 Gly distance is the largest in the Session 96 paper series.

■# PAPER*349 — SPT-CL J2215: Highest F*U*Bi*i in UQFF Dataset — Cool Core Starburst at z=1.16

Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** HIGHEST F*U*Bi_i in UQFF catalog; FIRST UQFF cool core starburst cluster at z>1 **Author:** Daniel T. Murphy

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

SPT-CL J2215-3537 (z = 1.16, M = 7.32×10^{1}■ M■, SFR ≈ 700 M■/yr) is the most extreme cool-core cluster in the South Pole Telescope sample and yields the highest UQFF buoyancy-unified force in the entire PAPER*346–352 dataset: F*UB*i*i ≈ -1.40×10^{21}■ N. The extreme starburst provides an independently measured SFR confirming the UQFF SFR = ρ*gas·v*wind·f*res formula. The x*2 = 8.4 Gly distance is the largest in the Session 96 paper series.

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2. Core Physics

2.1 UQFF Buoyancy-Unified Force (HIGHEST VALUE)

```
F_{U\_Bi\_i} \approx -1.40 \times 10^{218}\ \mathrm{N}
```

This exceeds the baseline AGN F*U*Bi_i = -8.32×10^{21}■ N by a factor of 1.68×, reflecting the enhanced vacuum buoyancy in an extreme cool-core environment.

2.2 Cool Core SFR — UQFF Prediction

The UQFF SFR:

```
\mathrm{SFR} = \rho_{\rm gas} \cdot v_{\rm wind} \cdot f_{\rm res}
```

Compared with observed SFR = 700 M■/yr. The UQFF calibration:

```
700\ M_\odot/\mathrm{yr} = \rho_{\rm gas}(r_{\rm cool}) \cdot v_{\rm turbulence} \cdot
f_{\rm TRZ}
```

2.3 Redshift-Scaled Separation

```
x_2 = c z / H_0 \cdot \chi(z) = 8.4\ \mathrm{Gly} = 7.95 \times 10^{25}\ \mathrm{m}
```

(comoving distance at z = 1.16, using Planck 2018 cosmology)

2.4 Cluster Mass Context

```
M_{\rm cl} = 7.32 \times 10^{14}\ M_\odot = 7.32 \times 10^{14} \times 1.989\times
10^{30}\ \mathrm{kg} = 1.46 \times 10^{45}\ \mathrm{kg}
```

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3. Key Values

Quantity	Formula	Value
z	Spectroscopic	1.16
M_cl	SPT mass	7.32×10^{1}■ M■
SFR	JCMT/ALMA obs	~700 M■/yr
F*U*Bi_i	UQFF full 5-eq	–1.40×10^{21}■ N
x_2	Comoving distance	8.4 Gly
F*U*Bi_i / baseline	Ratio to PAPER_346	×1.68

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4. Physical Significance

SPT-CL J2215 is the landmark test for UQFF at the highest-redshift cool-core + starburst intersection. The factor-of-1.68 enhancement in F*U*Bi*i above the baseline (–8.32×10^{21}■ N) provides the first quantitative UQFF prediction for why extreme cool-core clusters exhibit anomalously high SFRs: the elevated vacuum buoyancy (higher* ρSCm/ρ_UA ratio in dense cool cores) directly amplifies the buoyancy force and hence the gas compression rate.

The z = 1.16 observation epoch corresponds to a lookback time of ~8 Gyr — when the Universe was 46% of its current age — confirming UQFF cool-core physics operate at cosmic noon.

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5. Deduplication Note

- **vs. PAPER_350 (El Gordo):** El Gordo also yields F_U_Bi_i ≈ –1.40×10^{21}■ N but from a different mechanism (high mass + high velocity merger vs. extreme SFR in cool core).
- **vs. all other PAPER_346–352:** SPT-CL J2215 is unique as the only cool-core starburst cluster in the series.

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6. Classification

Physics Territory: HIGHEST UQFF F*U*Bi_i in dataset; FIRST cool-core starburst cluster at z > 1 **Scale:** Cosmological (M ~ 10^{1}■ M■, z = 1.16) **CP Implementation:** `SPTClJ2215CoolCoreStarburstCalculator` (CondensedPhysics3.py, Session 96)
